

Q1 2021 (01 Sep Shift 2)

The range of the function,

$$f(x) = \log_{\sqrt{5}} \left(3 + \cos\left(\frac{3\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} - x\right) - \cos\left(\frac{3\pi}{4} - x\right) \right) \text{ is:}$$

(1) $(0, \sqrt{5})$

(2) $[-2, 2]$

(3) $\left[\frac{1}{\sqrt{5}}, \sqrt{5}\right]$

(4) $[0, 2]$

Q2 2021 (26 Aug Shift 2)

The value of

$$2 \sin\left(\frac{\pi}{8}\right) \sin\left(\frac{2\pi}{8}\right) \sin\left(\frac{3\pi}{8}\right) \sin\left(\frac{5\pi}{8}\right) \sin\left(\frac{6\pi}{8}\right) \sin\left(\frac{7\pi}{8}\right)$$

is:

(1) $\frac{1}{4\sqrt{2}}$

(2) $\frac{1}{4}$

(3) $\frac{1}{8}$

(4) $\frac{1}{8\sqrt{2}}$

Answer Key

Q1 (4)

Q2 (3)

Q1 (4)

$$f(x) = \log_{\sqrt{5}}$$

$$\left(3 + \cos\left(\frac{3\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} - x\right) - \cos\left(\frac{3\pi}{4} - x\right)\right)$$

$$f(x) = \log_{\sqrt{3}} \left[3 + 2 \cos\left(\frac{\pi}{4}\right) \cos(x) - 2 \sin\left(\frac{3\pi}{4}\right) \sin(x) \right]$$

$$f(x) = \log_{\sqrt{5}} [3 + \sqrt{2}(\cos x - \sin x)]$$

$$\text{Since } -\sqrt{2} \leq \cos x - \sin x \leq \sqrt{2}$$

$$\Rightarrow \log_{\sqrt{5}} [3 + \sqrt{2}(-\sqrt{2})] \leq f(x) \leq \log_{\sqrt{5}} [3 + \sqrt{2}(\sqrt{2})]$$

$$\Rightarrow \log_{\sqrt{5}}(1) \leq f(x) \leq \log_{\sqrt{5}}(5)$$

So Range of $f(x)$ is $[0, 2]$

Option (4)

Q2 (3)

$$2 \sin\left(\frac{\pi}{8}\right) \sin\left(\frac{2\pi}{8}\right) \sin\left(\frac{3\pi}{8}\right) \sin\left(\frac{5\pi}{8}\right) \sin\left(\frac{6\pi}{8}\right) \sin\left(\frac{7\pi}{8}\right)$$

$$2 \sin^2 \frac{\pi}{8} \sin^2 \frac{2\pi}{8} \sin^2 \frac{3\pi}{8}$$

$$\sin^2 \frac{\pi}{8} \sin^2 \frac{3\pi}{8}$$

$$\sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8}$$

$$\frac{1}{4} \sin^2\left(\frac{\pi}{4}\right) = \frac{1}{8}$$